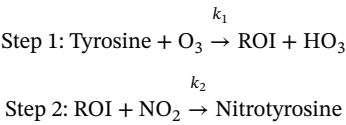


In polluted urban environments, airborne proteins can undergo nitration reactions that increase their allergenic potential toward the human respiratory system. For example, upon exposure to ozone (O₃) and nitrogen dioxide (NO₂) under atmospheric conditions, the amino acid tyrosine is known to undergo a two-step irreversible reaction leading to the formation of nitrotyrosine. The first step involves the formation of a reactive oxygen intermediate (ROI) tyrosyl radical through oxidation of the phenolic site of tyrosine. Nitrotyrosine is then formed in the second step through addition of NO₂ by a radical-radical termination reaction. The elementary steps for this reaction are shown in Scheme 1.



Scheme 1 Elementary steps of nitration of tyrosine

Airborne protein particles can exist in a variety of environments and in various phases including the aqueous phase, gaseous phase, or as conglomerates with other biomolecules such as lipids. Because the thermodynamic properties and kinetics of the nitration reaction vary based on the chemical environment in which it occurs, recent studies have focused on understanding the reaction under different conditions.

Density functional theory is a powerful computational quantum-mechanical modeling method that can determine the properties of many-electron systems by producing approximate solutions to the Schrödinger equation. Researchers used density functional theory to model the geometries and potential energy surfaces of the reactants, intermediates, and products of the nitration reaction in gaseous, aqueous, and lipid-rich environments. From this modeling, the thermodynamic and kinetic properties were calculated. Figure 1 shows the calculated activation energies *E_a* in kJ/mol, and Table 1 lists the associated thermochemical properties for both steps of the reaction in all three environments.

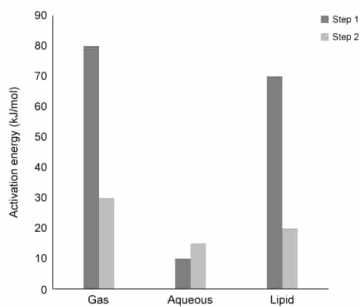
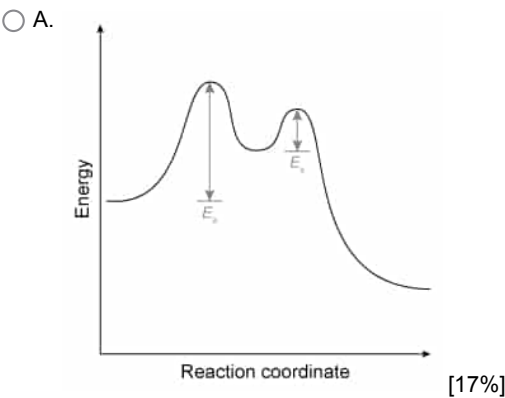


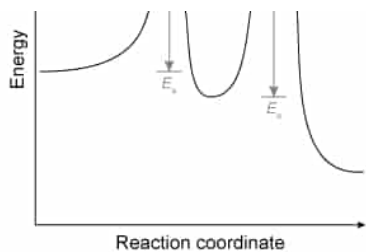
Figure 1 Calculated activation energies for each step of the studied reactions in three environments

Table 1 Calculated Thermochemical Properties of the Studied Reactions (298 K)

	Gas			Aqueous			Lipid		
Property (kJ/mol)	ΔH	ΔS	ΔG	ΔH	ΔS	ΔG	ΔH	ΔS	ΔG
Step 1	-90.9	-0.0159	-95.7	-18.4	-0.0337	-85.9	-104.3	-0.0415	-91.8
Step 2	0.5	-0.0001	0.9	-0.0045	-0.0045	-13.1	-15.3	-0.0042	-14.1

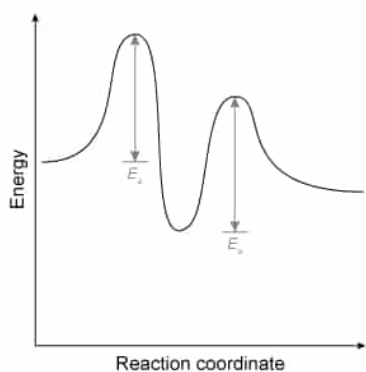
Which of the following reaction energy diagrams best represents the two-step reaction involving the conversion of tyrosine to nitrotyrosine in a lipid environment?





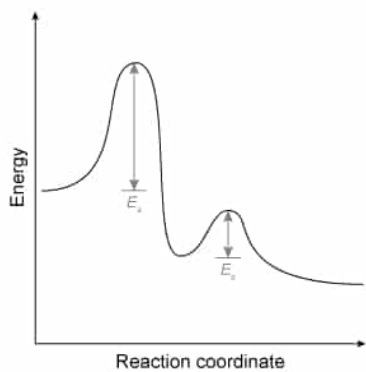
[3%]

☐ C.



[4%]

☒ D.

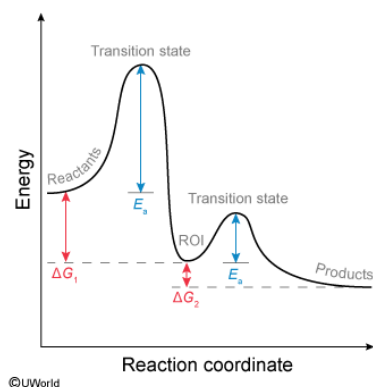


[73%]

Correct

73% answered correctly

Explanation:



A **reaction energy diagram** shows how the **energy of a system** changes during a **chemical reaction**. The graph consists of **Gibbs free energy** G on the y-axis and the **reaction coordinate** (ie, reaction progress) on the x-axis, with reactants on the left and products on the right. The energy difference between the reactants and the products is the change in Gibbs free energy ΔG :

$$\Delta G = G_{\text{products}} - G_{\text{reactants}}$$

If the reactants are higher in energy than the products, ΔG is **negative**, which indicates that the reaction is **spontaneous**. Conversely, if the reactants are lower in energy than the products, ΔG is **positive**, which indicates that the reaction is **nonspontaneous**.

The **activation energy** E_a (ie, the minimum amount of energy required to initiate a reaction) is the difference between the free energy of the **transition state** and the energy of the reactants (or **intermediates**). Because E_a is the energy required by the reactants to form a higher energy transition state, E_a is always positive and seen as a peak on the curve. A multistep reaction contains the same number of peaks as there are steps.

Based on the passage, the nitration of tyrosine in a lipid environment is a two-step reaction (Scheme 1). According to Figure 1 and

Table 1, the first step has a large activation energy with a ΔG of -91.8 kJ/mol and the second step has a smaller activation energy with a ΔG of -14.1 kJ/mol. Therefore, the reaction energy diagram depicted in Choice D best represents the energetics of the nitration reaction.

(Choice A) The Gibbs free energy ΔG of Step 1 is shown as positive but should be negative, according to Table 1.

(Choice B) The activation energy for Step 2 is shown as greater than the activation energy for Step 1 but should be less, according to Figure 1.

(Choice C) The Gibbs free energy ΔG for Step 2 is shown as positive but should be negative, according to Table 1.

Educational objective:

The reaction coordinate diagram is a representation of the change in energy of a system during a chemical reaction. The relative energies of the reactants, transition states, intermediates, and products are illustrated as their respective heights on the diagram.

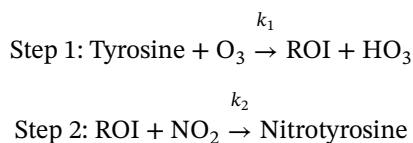
Time spent: 0 sec

Subject:
General Chemistry

QID: 400329

System:
Thermodynamics, Kinetics & Gas Laws

In polluted urban environments, airborne proteins can undergo nitration reactions that increase their allergenic potential toward the human respiratory system. For example, upon exposure to ozone (O₃) and nitrogen dioxide (NO₂) under atmospheric conditions, the amino acid tyrosine is known to undergo a two-step irreversible reaction leading to the formation of nitrotyrosine. The first step involves the formation of a reactive oxygen intermediate (ROI) tyrosyl radical through oxidation of the phenolic site of tyrosine. Nitrotyrosine is then formed in the second step through addition of NO₂ by a radical-radical termination reaction. The elementary steps for this reaction are shown in Scheme 1.



Scheme 1 Elementary steps of nitration of tyrosine

Airborne protein particles can exist in a variety of environments and in various phases including the aqueous phase, gaseous phase, or as conglomerates with other biomolecules such as lipids. Because the thermodynamic properties and kinetics of the nitration reaction vary based on the chemical environment in which it occurs, recent studies have focused on understanding the reaction under different conditions.

Density functional theory is a powerful computational quantum-mechanical modeling method that can determine the properties of many-electron systems by producing approximate solutions to the Schrödinger equation. Researchers used density functional theory to model the geometries and potential energy surfaces of the reactants, intermediates, and products of the nitration reaction in gaseous, aqueous, and lipid-rich environments. From this modeling, the thermodynamic and kinetic properties were calculated. Figure 1 shows the calculated activation energies E_a in kJ/mol, and Table 1 lists the associated thermochemical properties for both steps of the reaction in all three environments.

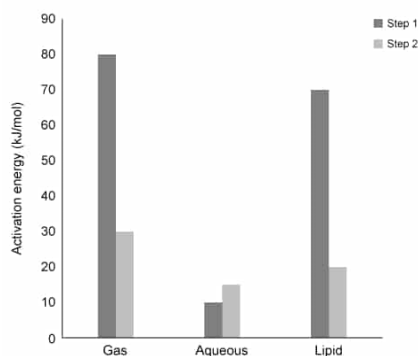


Figure 1 Calculated activation energies for each step of the studied reactions in three environments

Table 1 Calculated Thermochemical Properties of the Studied Reactions (298 K)

	Gas			Aqueous			Lipid		
Property (kJ/mol)	ΔH	ΔS	ΔG	ΔH	ΔS	ΔG	ΔH	ΔS	ΔG
Step 1	-90.9	-0.0159	-95.7	-18.4	-0.0337	-85.9	-104.3	-0.0415	-91.8
Step 2	0.5	-0.0001	0.9	-0.0045	-0.0045	-13.1	-15.3	-0.0042	-14.1

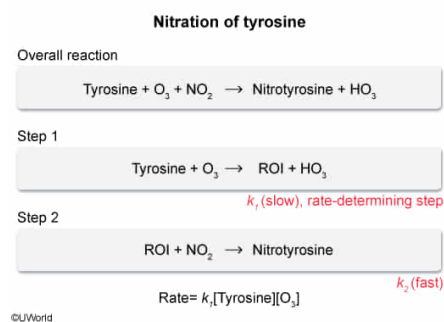
If the first step of the nitration of tyrosine in a lipid environment is slower than the second step, what is the rate law for the overall reaction?

- ✔ ☒ A. Rate = $k_1[\text{tyrosine}][\text{O}_3]$ [78%]
☐ B. Rate = $k_1[\text{tyrosine}][\text{O}_3][\text{NO}_2]$ [11%]
☐ C. Rate = $k_2[\text{ROI}][\text{NO}_2]$ [6%]
☐ D. Rate = $k_2[\text{tyrosine}][\text{O}_3]$ [3%]

Correct

79% answered correctly

Explanation:



The **rate law** of a chemical reaction is a mathematical equation that relates the **reaction rate** to the **reactant concentrations**, each raised to an experimentally determined exponent and multiplied by a rate constant k . The **reaction order** is equal to the sum of the exponents for each reactant in the rate law. When no exponent is written next to the reactant concentration, it is implied that the exponent is equal to 1.

For **elementary reactions** (ie, one-step reactions, individual steps of a multistep reaction), the **rate law** is expressed as the product of the rate constant multiplied by the concentration of the reactants, each raised to the power of its **stoichiometric coefficient**. For a reaction that requires more than one step, the **slowest** elementary step (ie, the **rate-determining step**) determines the rate law for the entire reaction. If the initial step is the slowest, the rate law is determined simply from that step using the coefficients of each reactant as exponents in the rate law. However, if the slowest step is the second step, **reaction intermediates** play a role in the rate law.

The nitration of tyrosine is a multistep reaction in which the first step is the rate-determining (slow) step. Therefore, the rate law can be written using the reactants of Step 1:

$$\text{Rate} = k_1[\text{tyrosine}][\text{O}_3]$$

(Choice B) This expression was written using reactants of the *overall* reaction instead of those of the rate-determining (slow) step.

(Choice C) The fast step of the reaction was used as the rate-determining step for this expression when the slow step should have been used.

(Choice D) The incorrect rate constant k_2 is used for this expression instead of the rate constant k_1 from the rate-determining step.

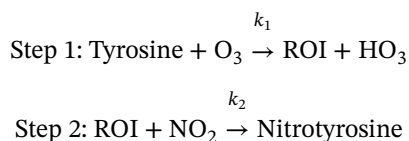
Educational objective:

Rate laws for molecular reactions relate the rate of reaction to the concentrations of reactants. For a multistep reaction, the rate law must be determined experimentally by determining the rate-limiting step (ie, the slowest step).

Time spent: 0 sec
Subject:
General Chemistry

QID: 400330
System:
Thermodynamics, Kinetics & Gas Laws

In polluted urban environments, airborne proteins can undergo nitration reactions that increase their allergenic potential toward the human respiratory system. For example, upon exposure to ozone (O₃) and nitrogen dioxide (NO₂) under atmospheric conditions, the amino acid tyrosine is known to undergo a two-step irreversible reaction leading to the formation of nitrotyrosine. The first step involves the formation of a reactive oxygen intermediate (ROI) tyrosyl radical through oxidation of the phenolic site of tyrosine. Nitrotyrosine is then formed in the second step through addition of NO₂ by a radical-radical termination reaction. The elementary steps for this reaction are shown in Scheme 1.



Scheme 1 Elementary steps of nitration of tyrosine

Airborne protein particles can exist in a variety of environments and in various phases including the aqueous phase, gaseous phase, or as conglomerates with other biomolecules such as lipids. Because the thermodynamic properties and kinetics of the nitration reaction vary based on the chemical environment in which it occurs, recent studies have focused on understanding the reaction under different conditions.

Density functional theory is a powerful computational quantum-mechanical modeling method that can determine the properties of many-electron systems by producing approximate solutions to the Schrödinger equation. Researchers used density functional theory to model the geometries and potential energy surfaces of the reactants, intermediates, and products of the nitration reaction in gaseous, aqueous, and lipid-rich environments. From this modeling, the thermodynamic and kinetic properties were calculated. Figure 1 shows the calculated activation energies E_a in kJ/mol, and Table 1 lists the associated thermochemical properties for both steps of the reaction in all three environments.

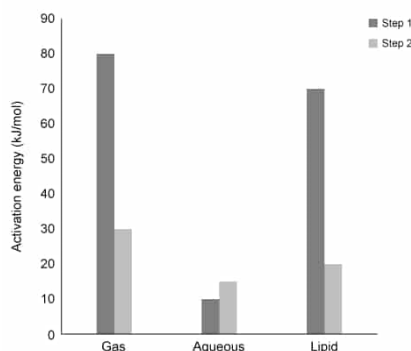


Figure 1 Calculated activation energies for each step of the studied reactions in three environments

Table 1 Calculated Thermochemical Properties of the Studied Reactions (298 K)

	Gas			Aqueous			Lipid		
Property (kJ/mol)	ΔH	ΔS	ΔG	ΔH	ΔS	ΔG	ΔH	ΔS	ΔG
Step 1	-90.9	-0.0159	-95.7	-18.4	-0.0337	-85.9	-104.3	-0.0415	-91.8
Step 2	0.5	-0.0001	0.9	-0.0045	-0.0045	-13.1	-15.3	-0.0042	-14.1

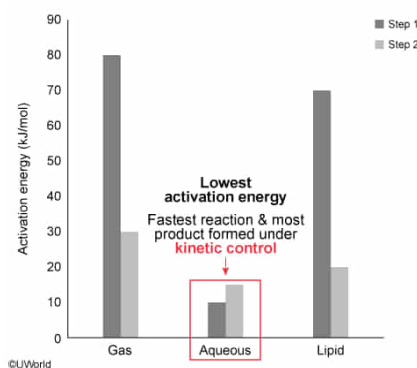
Suppose a chemist decides to conduct and monitor the reaction for 10 minutes at 298 K in the three environments described in the passage. Which of the three reactions is most likely to form the greatest amount of product over the 10-minute interval? (Assume the reaction has not reached completion.)

- ☐ A. The reaction in a lipid environment [17%]
- ✓ ☒ B. The reaction in an aqueous environment [64%]
- ☐ C. The reaction in a gaseous environment [11%]
- ☐ D. Cannot be determined from the information given [6%]

Correct

64% answered correctly

Explanation:



A reaction can be under either kinetic or thermodynamic control. When the conditions are such that a reaction is **irreversible** (eg, low temperature), it is said to be under **kinetic control**, and the most favorable product is the one whose pathway has the **lowest activation energy** E_a . The lower the E_a , the faster the reaction proceeds because a greater fraction of reactant molecules have enough **kinetic energy** to participate in **productive collisions** and overcome the E_a barrier.

When a reaction occurs under conditions such that it is **reversible** (eg, high temperatures), it is said to be under **thermodynamic control**. Under these conditions, all products form readily and the **most stable product** is **avored** (ie, the most **negative ΔG**), as it is least likely to undergo the reverse reaction.

According to the passage, the reactions are all irreversible at 298 K, and therefore all are under kinetic control. As a result, the reaction in aqueous solution will likely produce the greatest amount of product over 10 minutes because it has the lowest activation energy and proceeds the fastest. Note that it is assumed the reactions do not reach completion over the 10-minute interval.

(Choice A) Although the product is most stable for the reaction in a lipid environment, this attribute is irrelevant for reactions under kinetic control as the reverse reaction is unfeasible. Furthermore, this reaction has a higher activation energy than the same reaction in an aqueous environment.

(Choice C) The reaction pathway in a gaseous environment has a very large activation energy and is less favorable under kinetic control.

(Choice D) Because the passage states the reactions are under irreversible conditions, it can be assumed they are under kinetic control. Knowledge of the activation energies for the three reactions is sufficient to determine the most favorable product.

Educational objective:

Irreversible reactions tend to be under kinetic control, which favors products whose pathways have the lowest activation energy. Reversible reactions tend to be under thermodynamic control, which favors the most stable product.

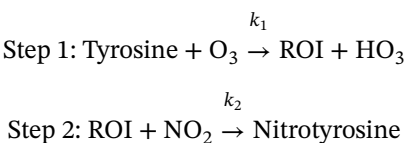
Time spent: 0 sec

Subject:
General Chemistry

QID: 400331

System:
Thermodynamics, Kinetics & Gas Laws

In polluted urban environments, airborne proteins can undergo nitration reactions that increase their allergenic potential toward the human respiratory system. For example, upon exposure to ozone (O₃) and nitrogen dioxide (NO₂) under atmospheric conditions, the amino acid tyrosine is known to undergo a two-step irreversible reaction leading to the formation of nitrotyrosine. The first step involves the formation of a reactive oxygen intermediate (ROI) tyrosyl radical through oxidation of the phenolic site of tyrosine. Nitrotyrosine is then formed in the second step through addition of NO₂ by a radical-radical termination reaction. The elementary steps for this reaction are shown in Scheme 1.



Scheme 1 Elementary steps of nitration of tyrosine

Airborne protein particles can exist in a variety of environments and in various phases including the aqueous phase, gaseous phase, or as conglomerates with other biomolecules such as lipids. Because the thermodynamic properties and kinetics of the nitration reaction vary based on the chemical environment in which it occurs, recent studies have focused on understanding the reaction under different conditions.

Density functional theory is a powerful computational quantum-mechanical modeling method that can determine the properties of many-electron systems by producing approximate solutions to the Schrödinger equation. Researchers used density functional theory to model the geometries and potential energy surfaces of the reactants, intermediates, and products of the nitration reaction in gaseous, aqueous, and lipid-rich environments. From this modeling, the thermodynamic and kinetic properties were calculated. Figure 1 shows the calculated activation energies *E_a* in kJ/mol, and Table 1 lists the associated thermochemical properties for both steps of the reaction in all three environments.

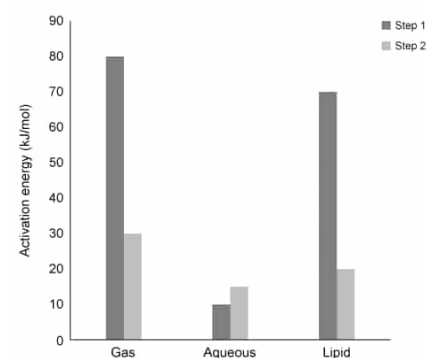


Figure 1 Calculated activation energies for each step of the studied reactions in three environments

Table 1 Calculated Thermochemical Properties of the Studied Reactions (298 K)

	Gas			Aqueous			Lipid		
Property (kJ/mol)	ΔH	ΔS	ΔG	ΔH	ΔS	ΔG	ΔH	ΔS	ΔG
Step 1	-90.9	-0.0159	-95.7	-18.4	-0.0337	-85.9	-104.3	-0.0415	-91.8
Step 2	0.5	-0.0001	0.9	-0.0045	-0.0045	-13.1	-15.3	-0.0042	-14.1

Given that entropy is negative (ie, disorder decreases) for the reaction, which of the following statements is true regarding the formation of aqueous nitrotyrosine?

- ☐ A. The formation of nitrotyrosine is less thermodynamically favorable than the formation of ROI at 298 K. [15%]
- ☐ B. The overall reaction becomes more thermodynamically favorable at higher temperatures. [22%]
- ☐ C. Neither the formation of ROI nor nitrotyrosine is thermodynamically favorable at 298 K. [5%]
- ☒ D. The overall reaction becomes less thermodynamically favorable at higher temperatures. [56%]

Correct

57% answered correctly

Explanation:

Effect of temperature on Gibbs free energy

$\Delta G = \Delta H - T\Delta S$

$\Delta G < 0$ spontaneous (thermodynamically favorable)
 $\Delta G > 0$ nonspontaneous (thermodynamically unfavorable)

ΔH	ΔS	$-T\Delta S$	ΔG with rising temperature	Spontaneity with rising temperature (T)
-	+	-	Increasingly negative	Always spontaneous
+	-	+	Increasingly positive	Always nonspontaneous
-	-	+	Decreasingly negative then increasingly positive	Spontaneous at low T; nonspontaneous at high T
+	+	-	Decreasingly positive then increasingly negative	Spontaneous at high T; nonspontaneous at low T

©UWorld

The **change in Gibbs free energy ΔG** of a reaction occurring at constant temperature and pressure is calculated from two **state functions**: the changes in **enthalpy ΔH** and **entropy ΔS** of a system. Gibbs free energy determines the reaction's **spontaneity**. A **spontaneous** reaction has a **negative** value of ΔG (ie, thermodynamically favorable) whereas a **nonspontaneous** reaction has a **positive** value of ΔG (ie, thermodynamically unfavorable). According to the equation

$$\Delta G = \Delta H - T\Delta S$$

the sign of ΔS determines whether a reaction becomes more or less thermodynamically favorable as the temperature increases. A **negative ΔS** causes ΔG to become **less favorable** (ie, less negative) with increasing temperature whereas a **positive ΔS** causes ΔG to become **more favorable** (ie, more negative) with increasing temperature.

According to Table 1, ΔH and ΔS for the reaction are both negative. Therefore, increasing the temperature leads to an increasingly greater (less negative) ΔG as shown by the above equation, making it less thermodynamically favorable at higher temperatures.

(Choice A) The formation of nitrotyrosine is *more* thermodynamically favorable than the formation of ROI at 298 K

because $\Delta G = -85.9$ kJ/mol for the formation of ROI (Step 1), but $\Delta G = -85.9 + (-13.1) = -99.0$ kJ/mol for the formation of nitrotyrosine (Step 1 + Step 2).

(Choice B) The reaction becomes *less* thermodynamically favorable at higher temperatures because ΔS is negative for this reaction, causing an increase in ΔG (less negative) as the temperature rises.

(Choice C) The production of *both* ROI and nitrotyrosine are thermodynamically favorable at 298 K because they both have negative ΔG terms (Table 1).

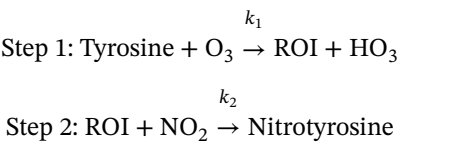
Educational objective:

The Gibbs free energy ΔG equation is used to determine the spontaneity of a system, where negative ΔG reflects a spontaneous reaction and a positive ΔG reflects a nonspontaneous reaction. The sign of the change in entropy ΔS for a process determines whether it becomes more or less thermodynamically favorable (ie, spontaneous) as the temperature changes.

Time spent: 0 sec
Subject:
General Chemistry

QID: 400332
System:
Thermodynamics, Kinetics & Gas Laws

In polluted urban environments, airborne proteins can undergo nitration reactions that increase their allergenic potential toward the human respiratory system. For example, upon exposure to ozone (O₃) and nitrogen dioxide (NO₂) under atmospheric conditions, the amino acid tyrosine is known to undergo a two-step irreversible reaction leading to the formation of nitrotyrosine. The first step involves the formation of a reactive oxygen intermediate (ROI) tyrosyl radical through oxidation of the phenolic site of tyrosine. Nitrotyrosine is then formed in the second step through addition of NO₂ by a radical-radical termination reaction. The elementary steps for this reaction are shown in Scheme 1.



Scheme 1 Elementary steps of nitration of tyrosine

Airborne protein particles can exist in a variety of environments and in various phases including the aqueous phase, gaseous phase, or as conglomerates with other biomolecules such as lipids. Because the thermodynamic properties and kinetics of the nitration reaction vary based on the chemical environment in which it occurs, recent studies have focused on understanding the reaction under different conditions.

Density functional theory is a powerful computational quantum-mechanical modeling method that can determine the properties of many-electron systems by producing approximate solutions to the Schrödinger equation. Researchers used density functional theory to model the geometries and potential energy surfaces of the reactants, intermediates, and products of the nitration reaction in gaseous, aqueous, and lipid-rich environments. From this modeling, the thermodynamic and kinetic properties were calculated. Figure 1 shows the calculated activation energies E_a in kJ/mol, and Table 1 lists the associated thermochemical properties for both steps of the reaction in all three environments.

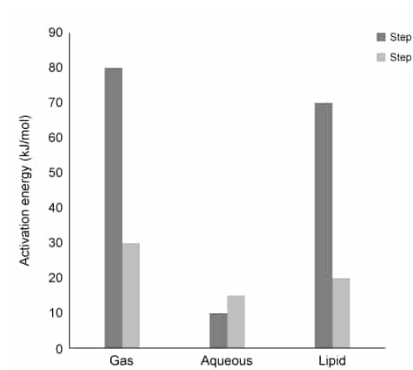


Figure 1 Calculated activation energies for each step of the studied reactions in three environments

Table 1 Calculated Thermochemical Properties of the Studied Reactions (298 K)

	Gas			Aqueous			Lipid		
Property (kJ/mol)	ΔH	ΔS	ΔG	ΔH	ΔS	ΔG	ΔH	ΔS	ΔG
Step 1	-90.9	-0.0159	-95.7	-18.4	-0.0337	-85.9	-104.3	-0.0415	-91.8
Step 2	0.5	-0.0001	0.9	-0.0045	-0.0045	-13.1	-15.3	-0.0042	-14.1

Which of the following are true concerning the overall reaction in an aqueous environment?

- I. The rate constant increases exponentially with increasing temperature.
- II. The rate constant increases exponentially with decreasing activation energy.
- III. The reaction rate decreases exponentially with increasing activation energy.
- IV. The rate constant of the catalyzed reaction is smaller than the rate constant of the uncatalyzed reaction.

- ☐ A. I and II only [18%]
- ✓ ☒ B. I, II, and III only [55%]
- ☐ C. II and IV only [13%]
- ☐ D. I, III, and IV only [12%]

Correct

55% answered correctly

Explanation:

Arrhenius equation

$$k = Ae^{\frac{-E_a}{RT}} = \frac{A}{e^{\frac{E_a}{RT}}}$$

	Reaction rate	Rate constant, k	Activation energy, E_a	Temperature, T
Number I		↑		↑
Number II		↑	↓	
Number III	↓	↓	↑	
Number IV		↓ uncatalyzed ↑ catalyzed	↑ ↓	

©UWorld

The **rate constant** k of a reaction is related to its **activation energy** E_a and **temperature**, according to the **Arrhenius equation**:

$$k = Ae^{-\frac{E_a}{RT}}$$

The negative exponential term in the equation can then be rewritten in the denominator:

$$k = \frac{A}{e^{\frac{E_a}{RT}}}$$

Increasing the temperature of a reaction yields a smaller value of the exponential term. Because it is in the denominator, a smaller exponential term gives an increased value for k (**Number I**).

Decreasing E_a also gives a smaller exponential term, and therefore k increases exponentially with decreasing E_a (**Number II**). Conversely, k decreases exponentially with increasing E_a . According to the **rate law equation**, the rate of a given reaction is directly proportional to the rate constant. Therefore, the reaction rate also decreases exponentially with increasing E_a (**Number III**).

(**Number IV**) A **catalyst** lowers the value of E_a of a reaction, and therefore an uncatalyzed reaction has a higher E_a than a catalyzed reaction. Because uncatalyzed reactions have a higher E_a , the reaction has a smaller rate constant with respect to the rate constant of the catalyzed reaction.

Educational objective:

The Arrhenius equation describes the relationship between the rate constant, activation energy, and temperature of a reaction. Both the rate constant and the rate of reaction decay exponentially with increasing activation energy and increase exponentially with increasing temperature. Because catalysts lower the activation energy of a reaction, the rate constant is greater for a catalyzed reaction than for an uncatalyzed reaction.

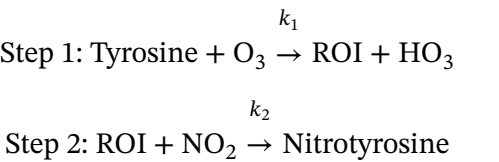
Time spent: 0 sec

Subject:
General Chemistry

QID: 400333

System:
Thermodynamics, Kinetics & Gas Laws

In polluted urban environments, airborne proteins can undergo nitration reactions that increase their allergenic potential toward the human respiratory system. For example, upon exposure to ozone (O₃) and nitrogen dioxide (NO₂) under atmospheric conditions, the amino acid tyrosine is known to undergo a two-step irreversible reaction leading to the formation of nitrotyrosine. The first step involves the formation of a reactive oxygen intermediate (ROI) tyrosyl radical through oxidation of the phenolic site of tyrosine. Nitrotyrosine is then formed in the second step through addition of NO₂ by a radical-radical termination reaction. The elementary steps for this reaction are shown in Scheme 1.



Scheme 1 Elementary steps of nitration of tyrosine

Airborne protein particles can exist in a variety of environments and in various phases including the aqueous phase, gaseous phase, or as conglomerates with other biomolecules such as lipids. Because the thermodynamic properties and kinetics of the nitration reaction vary based on the chemical environment in which it occurs, recent studies have focused on understanding the reaction under different conditions.

Density functional theory is a powerful computational quantum-mechanical modeling method that can determine the properties of many-electron systems by producing approximate solutions to the Schrödinger equation. Researchers used density functional theory to model the geometries and potential energy surfaces of the reactants, intermediates, and products of the nitration reaction in gaseous, aqueous, and lipid-rich environments. From this modeling, the thermodynamic and kinetic properties were calculated. Figure 1 shows the calculated activation energies *E_a* in kJ/mol, and Table 1 lists the associated thermochemical properties for both steps of the reaction in all three environments.

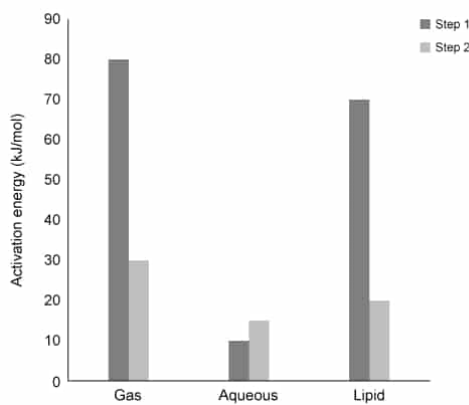
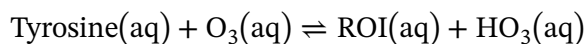


Figure 1 Calculated activation energies for each step of the studied reactions in three environments

Table 1 Calculated Thermochemical Properties of the Studied Reactions (298 K)

	Gas			Aqueous			Lipid		
Property (kJ/mol)	ΔH	ΔS	ΔG	ΔH	ΔS	ΔG	ΔH	ΔS	ΔG
Step 1	-90.9	-0.0159	-95.7	-18.4	-0.0337	-85.9	-104.3	-0.0415	-91.8
Step 2	0.5	-0.0001	0.9	-0.0045	-0.0045	-13.1	-15.3	-0.0042	-14.1

Because Step 2 of the reaction in aqueous phase is rate-limiting, the formation of ROI from tyrosine and ozone in Step 1 can be assumed to reach a state of equilibrium:



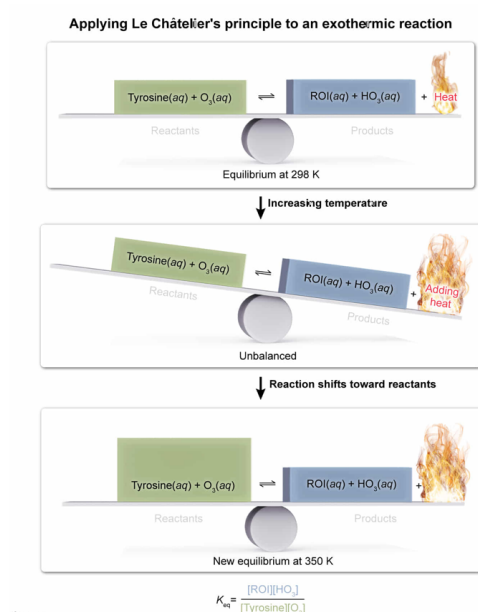
Which of the following will occur if the temperature of this system is raised from 298 K to 350 K?

- ✓ ☒ A. K_{eq} will decrease as the reaction is driven toward reactants. [45%]
☐ B. K_{eq} will increase as the reaction is driven toward products. [35%]
☐ C. K_{eq} will decrease as the reaction is driven toward products. [8%]
☐ D. K_{eq} will remain constant as the reaction is driven toward reactants. [10%]

Correct

45% answered correctly

Explanation:

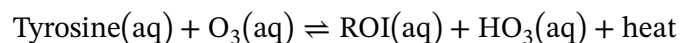


Equilibrium is reached in a reaction when the rate at which products are formed is equal to the rate at which reactants are formed. A reaction's **equilibrium constant** K_{eq} indicates to what degree reactants are converted to products and is equal to the equilibrium molar ratios of products to reactants:

$$K_{\text{eq}} = \frac{[\text{Products}]^{\text{coefficients}}}{[\text{Reactants}]^{\text{coefficients}}}$$

The **temperature dependence** of K_{eq} depends on whether a reaction is **exothermic** ($\Delta H^\circ < 0$) or **endothermic** ($\Delta H^\circ > 0$). An exothermic reaction releases heat (ie, heat is treated as a product), and **increasing** the temperature (ie, adding heat) causes the reaction to shift toward the reactants (ie, K_{eq} decreases) to compensate for the heat gain in accordance with **Le Châtelier's principle**. Conversely, an endothermic reaction absorbs heat (ie, heat is treated as a reactant), and the reaction shifts toward products (ie, K_{eq} increases) with **increasing** temperature.

According to Table 1, the reaction between tyrosine and O_3 (Step 1) in an aqueous environment is exothermic ($\Delta H = -18.4 \text{ kJ/mol}$). As such, heat can be viewed as a product of the reaction:



Raising the temperature (ie, adding heat) causes the reaction to shift toward the reactants to rebalance the equilibrium reaction. The shift toward reactants leads to an increase in reactant concentration, and therefore a decrease in K_{eq} .

(Choices B, C, and D) Because the reaction is *exothermic*, raising the temperature causes the reaction to be driven toward the *reactants*, increasing their concentration and causing a *decrease* in K_{eq} rather than an increase or remaining constant.

Educational objective:

Equilibrium is reached in a reaction when the rate at which products are formed is equal to the rate at which reactants are formed. The equilibrium constant K_{eq} is defined as the ratio of products to reactants when a reaction is in equilibrium; only changes in temperature can alter K_{eq} . Le Châtelier's principle states that a system responds to disturbances in equilibrium by shifting the reaction to counter the effect of the disturbance.

Time spent:0 sec

Subject:
General Chemistry

QID:400334

System:
Thermodynamics, Kinetics & Gas Laws